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Neural Networks in Volatility Forecasting: A
comparative analysis of GARCH and neural networks on
the Swedish, Polish and Slovenian stock indexes

Neurala nätverk inom volatilitetsprognoser: En jämförande analys
på GARCH och neurala nätverk på svenska, polska och slovenska
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Abstract

In recent years, the interest in machine learning techniques has exploded with widespread applications in many fields. In this study, we examine the effectiveness of neural networks in the context of financial market volatility forecasting and how they perform relative to econometric volatility forecasting models. To do this, we apply these models on the Swedish index OMXSPI, Polish index WIG and Slovenian index SBITOP. We perform volatility forecasting using a wide range of GARCH family models and neural networks with an LSTM architecture and predict the forward 14-day volatility of daily returns using data from the 31st of July 2017 to the 31st of July 2023, with the last 342 days used for validation and testing. The in-sample performance and parameter selection of the econometric models were evaluated using Akaike information criterion and Bayesian information criterion metrics and the out-of-sample evaluation was done using root-mean-square percentage error (RMSPE) and root-mean-square error (RMSE). In addition, the GARCH model was implemented using both a sliding and expanding window, where the expanding window was superior in the context of RMSPE and RMSE on the validation data.

The experiments revealed that LSTM neural networks trained on trading data were consistently more effective in volatility forecasting based on their RMSPE and RMSE, and can therefore serve as a competitive method for volatility forecasting compared to GARCH-family econometric forecasting models.

Keywords: volatility forecasting, GARCH, neural networks

Sammanfattning

Under senare år har intresset för maskininlärning ökat explosionsartat med utbredd tillämpning i många områden. I denna studie undersöker vi effektiviteten hos neurala nätverk i samband med volatilitetsprognoser för finansiella marknader och hur de presterar i förhållande till ekonometriska volatilitetsmodeller. För att göra detta tillämpar vi dessa modeller på svenska aktieindexet OMXSPI, polska WIG och slovenska SBITOP. Vi utför volatilitetsprognoser med hjälp av ett brett utbud av GARCH-modeller och neurala nätverk av LSTM-arkitektur för att förutspå volatiliteten 14 dagar framåt hos de dagliga avkastningarna med hjälp av data från den 31 juli 2017 till 31 juli 2023 där de sista 342 dagarna används för validering och test av modellerna. De ekonometriska modellernas prestanda i provet utvärderades med hjälp av Akaike information criterion och Bayesian information criterion, och utvärderingen utanför provet gjordes med hjälp av root-mean-square percentage error (RMSPE) och root-mean-square error (RMSE). GARCH-modellerna implementerades också med både glidande och expanderande fönster, där ett expanderande fönster var överlägset sett till RMSPE och RMSE på valideringsdatan.

Experimenten visade att neurala nätverk tränade på handelsdata var konsekvent mer effektiva för volatilitetsprognoser hos OMXSPI, WIG och SBITOP baserat på RMSPE och RMSE, och kan således användas som en konkurrenskraftig metod för volatilitetsprognoser jämfört med de GARCH-modeller som använts.

Nyckelord: volatility forecasting, GARCH, neural networks

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1. Introduction

1.1 Introduction and problem formulation

“Lessons of days gone by teach us what will come to pass.”

– The Prior of Ver Eger, *Stargate*

Volatility is a central topic in finance and financial markets and it plays a pivotal role in pricing models, risk management, and derivative pricing. Changes in volatility trends are closely looked at by participants in the financial markets as changes in current and forecasted volatility influence investments and risk management decisions (Gregoriou 2009). In the financial industry, the distribution of returns is examined and subsequently used in forecasts in order to model and predict the risk that the financial institution is exposed to, now and in the future. History has taught us that the stability of financial institutions is crucial to a healthy economy and the benefits and necessity of volatility forecasting become clear.

Since the early 80s, many attempts to create models that can accurately forecast volatility have been made, most notably the GARCH model put forth by Bollerslev in 1986, based on the ARCH model by Engle. Subsequent additions to the GARCH model have been proposed in order to address some properties of financial market volatility such as asymmetric volatility (Zakoian 1994) and also model specific properties such as the non-negativity constraints of the parameters in ordinary GARCH models (Nelson 1991). In recent decades, major improvements to computation and accessibility to more complex modeling techniques have been achieved which has opened the door for models that can identify non-linear relationships in data which in turn enables vast applicability in many fields (Chow 2007). Neural networks have subsequently become a potential new alternative to the older econometric models used in the field of economics and finance and will be examined in the context of volatility forecasting in this thesis.

1.2 Purpose of the thesis

The purpose of this thesis is to address whether a neural network can yield a better volatility forecasting model than widely used econometric approaches. In detail, the study will answer the following question

- How effective are neural networks in comparison to GARCH-family econometric models in modeling financial market volatility?

1.3 Method

The data used for the study are the daily open, close, low, high, and volume for three European indexes, the OMXSPI, WIG, and SBITOP. The GARCH-family econometric models are estimated using the daily returns and chosen based on the in-sample performance metrics Akaike information criterion (AIC) and Bayesian information criterion (BIC), and the neural networks are trained on all the previously mentioned data. The performance of the models is evaluated using the cost functions root-mean-square error and root-mean-square percentage error and are at first measured on a validation dataset for model selection before being tested on the final test set.

1.4 Disposition

The disposition of the thesis is as follows: Chapter 2 briefly explains the history of volatility forecasting and the rationale behind the neural networks and econometric models. It also covers previous research on the topic of volatility forecasting. Chapter 3 covers the mathematics of neural networks in detail and how they may prove beneficial in volatility forecasting as opposed to older econometric models and the mathematics behind the econometric models covered in Chapter 2. Chapter 4 goes through the data in detail and covers the network and models that will be used for the quantitative analysis. Chapter 5 contains the results of the models, and the thesis concludes with Chapter 6 which covers the conclusion that can be drawn from the empirical analysis concerning the purpose of the thesis.

1.5 Delimitations

A delimitation of the paper is the access to computational power. To decide on a statistically "correct" window size for the sliding forecasting, one would have to do AIC and BIC tests

for each sliding size and for every model which is not practically possible. Instead, a static window size is used and the in-sample evaluation is done on the different number of model parameters to find the most appropriate parameter counts for each model.

2. Background

2.1 Neural networks

Neural networks, or more formally known as *artificial* neural networks were proposed in the early 1940s by McCullough and Pitts. The first neural network implementation done in practice occurred in the 50s by the American psychologist Frank Rosenblatt, often referred to as the "father of neural networks" with his *perceptron* (McCullough & Pitts 1943; Rosenblatt 1958; Tappert 2019).

The neural network is inspired by the brain, with each neuron being connected to other neurons by synapses, called *edges* in the context of artificial neural networks. The signals in the brain are subsequently the data that flows between the neurons, and the weights in the networks correspond to the strength of the signal. The way such a network learns is by comparing what it believes to be the closest answer to what the actual answer is. It then changes the steps that get it to the answer it thinks is correct and tries again. This process is repeated until it finds a way that minimizes the amount of faulty answers it gives. The mathematical interpretation of this concept is described in a following chapter.

2.2 Volatility

Volatility is defined as the standard deviation (σ) of the return of an asset and is a measure of how spread returns are across a certain time frame of a security or index (Hull, 2018). Financial market volatility is influenced by many different types of factors such as macroeconomic measures, for example GDP growth, but also factors such as oil prices, the financialization of commodities and financial market liberalization (Li et al. 2022). Two widely used quantifications of portfolio risk, namely *Value-at-Risk* (*VaR*) and *Expected Shortfall* (*ES*), use volatility as a risk measure, with high volatility leading to high VaR and ES, and vice versa (Hull 2018). Volatility is also used in the metric Sharpe ratio where the excess return is put

in relation to the volatility of the asset or portfolio, yielding a risk-adjusted return metric (Bodie et al. 2009).

2.3 Value-at-Risk

Value-at-Risk (VaR) is a metric used in risk management to measure potential losses. Specifically, it is a measurement of how much a portfolio can be expected to lose over a given time frame and with a certain confidence level. If a portfolio has a daily VaR of €4 million, at a 95% confidence level, the losses on any given day will only be higher than €4 million on average once every 20 trading days. The metric is therefore dependent on volatility to determine the probability distribution of the returns. A part of the Basel II and III regulatory framework is a set of standards on financial institutions' internal models for minimum capital requirements, involving Value-at-Risk. Furthermore, Value-at-risk metrics are used in the context of stress testing, involving the computation of VaR during financial stress. Volatility forecasting models can therefore be used to predict the expected Value-at-Risk and protect the financial institution from risk (Crouhy et al. 2014).

2.4 Expected shortfall

Similar to VaR , *Expected shortfall*, commonly abbreviated as ES , is used as a metric to quantify potential portfolio losses. It is similar to VaR in the sense that it models undesirable outcomes, however it deals with one weakness of VaR . VaR is a single value that measures *at least* how bad it can get. It however does not take into consideration the shape of the tail beyond that point, meaning that it does not work well if the potential losses beyond the confidence interval far exceed the value at the confidence interval. ES deals with this flaw by measuring the mean of all outcomes at and beyond the VaR measure. Aside from this difference, both ES and VaR require a confidence interval, time horizon, and a measurement of the volatility of the portfolio where volatility forecasting once again comes into the picture (Hull 2018).

2.5 The GARCH family

The model that gave rise to the GARCH family of volatility models was called the ARCH (Autoregressive conditional heteroscedasticity) model and became the basis of many other models such as GARCH (Generalized autoregressive heteroskedasticity model) and its subsequent

successors. They're based on the notion of intertemporal heteroscedasticity and volatility clustering and as a result, attempt to model the volatility as a function of past residuals and conditional volatilities, with additions such as dummy variables to handle asymmetry such as in the TGARCH and GJR-GARCH models, and logarithmic volatility modeling with the EGARCH model which allowed for the removal of non-negativity constraints. The GARCH family of forecasting models all have an arbitrary number of parameters. It is therefore required to determine the parameter count quantitatively such as with AIC or BIC. A brief overview of the purpose of the different GARCH models is given in the following subchapters.

2.5.1 Autoregressive conditional heteroskedasticity (ARCH)

The autoregressive conditional heteroscedasticity model was proposed by Engle in 1982 to model variance as a regression of previous residuals following a certain mean (Engle, 1982). The reason for its invention was the rather controversial practice of assuming homoscedastic variance in time series amongst economists. McNees claims that the variance does vary as a function of time, reinforcing the legitimacy of the ARCH model. Furthermore, McNees also emphasizes the idea that variance is connected over time as they tend to form clusters, where times of high volatility are a good indicator of high volatility in the near future, and vice versa for low volatility (McNees 1979, as cited in Engle 1982).

2.5.2 Generalized autoregressive conditional heteroscedasticity (GARCH)

The GARCH model was proposed in 1986 by Bollerslev and is an extension of the ARCH model brought forward by Engle in 1982. The GARCH model extends the ARCH model by introducing another variable aside from past residuals, namely the past conditional volatilities (Bollerslev 1986).

2.5.3 Exponential GARCH

While GARCH and ARCH assume that volatility is symmetrical, the EGARCH model handles volatility as if it is asymmetrical, meaning that negative and positive shocks to returns affect the volatility differently. Furthermore, it has no non-negativity constraints and it also captures volatility persistence (Nelson 1991).

2.5.4 Threshold GARCH

To tackle the issue of asymmetric volatility where the volatility increases disproportionately in a market downturn compared to a rise, Zakoïan developed upon the GARCH and extended it with a dummy variable that depends on the sign of the residual of the previous time-step. This model can also be seen as a variant of the GJR-GARCH model with the variance substituted for standard deviation (Zakoïan 1994).

2.5.5 GJR-GARCH

The GJR-GARCH model is similar to the TGARCH model in the sense that it directly tries to deal with the problems of modeling asymmetric volatility. It does this by observing the sign of the past residual, adding an extra factor on the sum if the previous residual was negative, causing an amplified effect on volatility if the market took a downturn the previous timestep (Glosten et al. 1993)

2.6 Literary review

There are several studies that suggest that neural networks are a viable alternative for volatility forecasting. (Bucci 2020) made a comparison between neural networks and autoregressive econometric models, namely LSTAR and ARFIMA, and found that LSTM and NARX neural networks were superior to these traditional econometric models. The neural networks were better both for one-step-ahead forecasts and forecasts over several timesteps. Furthermore, (Hansson 2017) found that LSTM networks are effective relative to ARMA(1,1) GJR-GARCH(1,1), however advises against trading strategies involving the technique. He also found that the predictability of the OMX, S&P500 and Bovespa varies, where OMX is more predictable than the others. He further explains that the ambiguity of neural networks makes it difficult to understand why they perform differently, but explains that one explanation could be that the training and test data simply share more similarities in the OMX sample compared to the training and test data of the Bovespa and S&P500.

When it comes to the topic of incorporating other variables into the GARCH model to further improve the models performance, there are three studies that lay the foundation for fine-tuning of the empirical analysis in this thesis. (Santos, et. al 2013) found that multivariate GARCH models perform comparatively better than univariate models on real and simulated data of portfolios. Furthermore, they found that a Student's t distribution and a

rolling window perform better than a fixed estimation window and other distributions. When it comes to the use of other variables, (Hung, et. al 2019) found that incorporating a market liquidity metric proposed by (Amihud 2002) into their models lead to an improvement in forecasting ability, which indicates that a measure for market liquidity would positively benefit the neural networks. Other proxies for volatility were explored in (Patton 2011) and (Liu, et. al 2012) which found that the intra-day range and realized volatility prove to be superior to noisier proxies such as squared returns and are therefore employed as proxies in the training and testing of the neural networks in this thesis.

This thesis complements the previous literature by incorporating more variations of the GARCH volatility models and explores more neural network architectures, and also provides more empirical evidence of choice of variables and forecasting technique, and brings forth further insight into GARCH- and neural network performance across different indexes based on market size and liquidity.

3. Theory

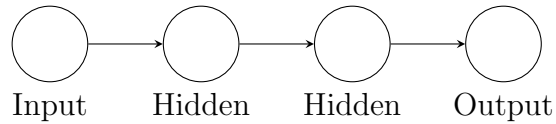
3.1 Neural networks in detail

A neural network is built up of three fundamental parts – artificial neurons (*neurons*), layers, and edges (*synapses*). This concept is inspired by the human brain, and modern artificial neural networks such as ChatGPT-4 contain over one trillion neurons (BigThink 2023). There are also several types of network architectures, tailored specifically to handle problems that arise with “basic” neural network design such as vanishing or exploding gradients when forecasting time series using neural networks. The two main networks that will be touched on briefly are *feed-forward networks* and *long-short term memory (LSTM)*.

3.1.1 Feed-forward network

Neural networks may be quite hard to grasp as they are complex and extensive by nature. There are also different kinds of network designs specialized for specific tasks. Therefore, entertain the following simple feed-forward neural network in Figure 3.1:

Figure 3.1: Single neuron layered neural network



The hidden neurons each represent a specific linear function that is passed into a transforming, non-linear function. The first hidden neuron in the hypothetical neural network takes the input data, x , and processes it with the function in Equation 3.1,

$$f(x) = w_1^{L-2}x + b_1^{L-2} \quad (3.1)$$

The output of the linear function is then transformed with a non-linearity, called an *activation*

function such as the following three commonly used functions, the *sigmoid*, the *rectified linear unit* and the *hyperbolic* function,

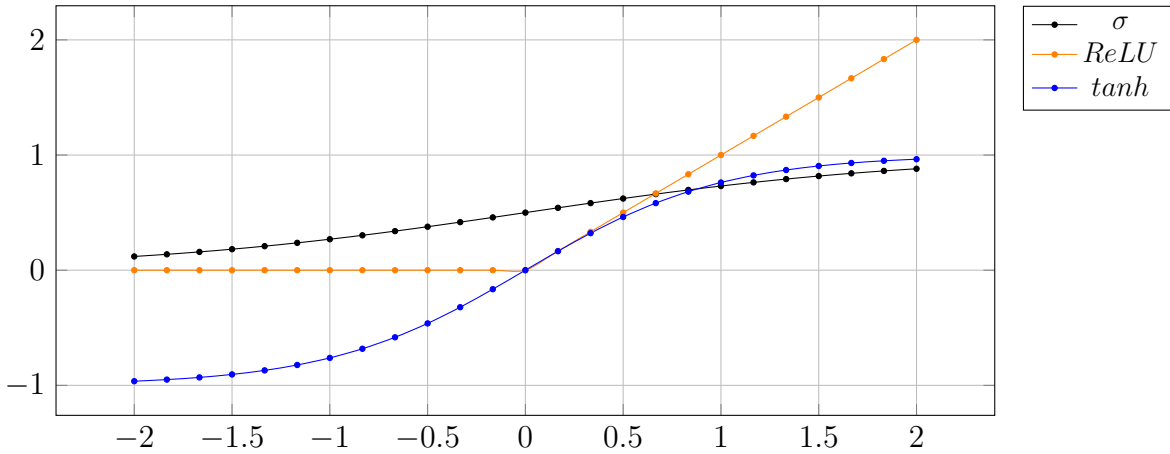
$$\sigma(f(x)) = \frac{1}{1 + e^{-f(x)}}$$

$$ReLU(f(x)) = \max(f(x), 0) \quad (3.2)$$

$$\tanh(f(x)) = \frac{e^{f(x)} - e^{-f(x)}}{e^{f(x)} + e^{-f(x)}}$$

Below in Figure 3.2 are the plots of the commonly used activation functions in neural networks.

Figure 3.2: The hyperbolic, the sigmoid and the ReLU functions



This is fundamentally where the non-linearity enters the neural network and allows it to identify non-linear patterns in data where other, less complex linear models would have less success.

More generally, you may compute the value of any given neuron, say neuron n in layer L where the previous layer had k neurons using the activation function f , in an arbitrarily complex network, as,

$$a_n^L = f\left(\sum_{i=1}^k w_i^{L-1} a_i^{L-1} + b_i^{L-1}\right) \quad (3.3)$$

where the sum to be transformed with the activation function is the weighted sum of all neurons from the previous layer. The final value of this neural network for a given input

may be denoted as a^L and is compared to the expected value from the output of the training data, y , often using a mean-square error or mean-absolute error cost function. More on the selection of cost function in an upcoming section. The goal is to minimize the chosen cost function for the training data set and use that model for forecasting, prediction, or other purposes. Finding the minimum for the cost function may be difficult computationally and while there are several ways to reduce the amount of computations needed, the most commonly used algorithm is a procedure called *back-propagation* which will be touched on briefly in an upcoming section (Aggarwal 2018).

3.1.2 Long-Short Term Memory (LSTM)

Originally proposed in 1996 by Hochreiter and Schmidhuber, the LSTM neural network was developed as an alternative to ordinary neural networks to deal with time series and data that is sequential by bridging different time intervals and preserving data across time. The problems with the “ordinary” neural network are referred to as the *vanishing gradient problem*, where the weight changes become very small to the point where they have no effect or exceedingly large, known as the *exploding gradient problem* (Hochreiter & Schmidhuber 1997). Each memory cell contains input weights, output weights, and an internal state. What makes an LSTM network special are the gates that handle the following:

- *Forget gate*: The decision on what information to “forget”.
- *Input gate*: The decision on what information from the input to be used to update the memory cell.
- *Output gate*: The decision on what to output given the input and the long-term and short-term memory cell.

The usage of an LSTM network as opposed to a feed-forward network provides the benefit of both solving the vanishing gradient and exploding gradient problems and therefore creates a framework for modeling large temporal dependencies in the input data (Brownlee 2017).

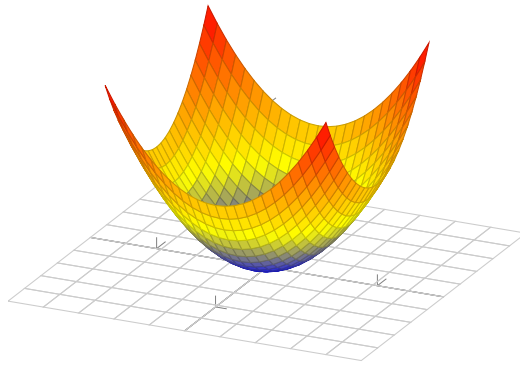
3.1.3 Back-propagation

Back-propagation is the most common algorithmic technique for the minimization of the cost function of a neural network. The goal is to compute the gradient of the cost function with respect to all the weights and biases in the network

$$\nabla C(w_1^1, b_1^1, w_2^1, \dots, w_K^L, b_K^L) = \begin{bmatrix} \frac{\partial C}{\partial w_1^1} \\ \frac{\partial C}{\partial b_1^1} \\ \frac{\partial C}{\partial w_2^1} \\ \vdots \\ \frac{\partial C}{\partial w_k^L} \\ \frac{\partial C}{\partial w_k^L} \end{bmatrix} \quad (3.4)$$

The gradient in Equation 3.4 is first computed by randomizing all the weights and biases in the network and then proceeding with the first data set. Using the computed gradient, the weights and biases in the neural network are adjusted to the opposite direction of the gradient incrementally to try to minimize the cost function. You may visualize this by Figure 3.3 below which is a two-variate model where the point on the surface we're trying to find is the lowest point relative to the vertical axis.

Figure 3.3: Visualization of cost function minimization using gradient descent



The way the gradient is computed is by using the chain rule. If one would want to find the partial derivative of the cost function, $C(\cdot)$, with respect to the weight w^L , then one must first take into consideration that the weight is first used in the linear function, call it z^L , and then transformed to a^L using some non-linear function and then finally, a^L is used in the cost function which is the function we're trying to derive. As mentioned above this can be computed using the chain rule such that:

$$\frac{\partial C}{\partial w^L} = \frac{\partial z^L}{\partial w^L} \frac{\partial a^L}{\partial z^L} \frac{\partial C}{\partial a^L} \quad (3.5)$$

Once the gradient has been computed, the weights and biases are changed in the network using the following function in Equation 3.6,

$$\theta_{n+1} = \theta_n - \eta \nabla_{\theta} C \quad (3.6)$$

where the η denotes the *learning rate*, the size of the steps we take in the optimization process and θ denotes any given bias or weight in the network. This process is repeated until the model converges towards a satisfactory final model, where the increments no longer have a meaningful effect on the minimization of the cost function.

3.2 The GARCH family

3.2.1 ARCH

The ARCH model brought forward by Engle assumes that volatility is temporal and attempts to model the variance by applying a regression on the previous residuals. This can be formalized as

$$\sigma_n^2 = \omega + \sum_{t=1}^m \alpha_t u_{n-t}^2 \quad (3.7)$$

where ω is the long-term volatility and the α terms represent the weights of the previous squared returns.

3.2.2 GARCH

As opposed to the ARCH model proposed by Engle, the GARCH model incorporates past conditional volatility as a predictor for future volatility. The model is denoted as GARCH(p,q), with the parameters indicating the number of past conditional volatilities and past residuals. The most widely used version of the GARCH model is the GARCH(1,1), using one term of the past squared residual and one term of the past variance such that

$$\sigma_t^2 = \omega + \alpha u_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (3.8)$$

where ω is the long term volatility, α is the coefficient of the ARCH term and β is the coefficient of the past variance estimates. However, as mentioned previously, GARCH may

use as many p and q terms, however the same procedure applies.

3.2.3 GJR-GARCH

GJR-GARCH deals with asymmetric volatility by adding a dummy variable. This dummy variable acts as an amplification to the forecast of the variance rate.

$$h_t = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \beta h_{t-1} + \gamma \epsilon_{t-1}^2 D_{t-1} \quad (3.9)$$

$$D_{t-1} = \begin{cases} 1, & \text{if } \epsilon_{t-1} < 0 \\ 0, & \text{otherwise.} \end{cases}$$

The dummy variable is denoted as D_{t-1} and works as an amplification by adding $\gamma \epsilon_{t-1}$ onto the conditional variance rate if the market took a downturn. By doing this, the model adjusts the forecast based on whether the previous time step had a positive or negative return and therefore takes into account asymmetry.

3.2.4 Threshold GARCH

Similar to GJR-GARCH, the threshold-GARCH model has a threshold variable to deal with asymmetric volatility.

$$\sqrt{h_t} = \alpha_0 + \alpha_1 |\epsilon_{t-1}| + \beta_1 \sqrt{h_{t-1}} + \gamma |\epsilon_{t-1}| I_{t-1} \quad (3.10)$$

$$I_{t-1} = \begin{cases} 1, & \text{if } \epsilon_{t-1} < 0 \\ 0, & \text{otherwise.} \end{cases}$$

It uses the lagged absolute residuals as opposed to the lagged squared residuals and lagged conditional standard deviations instead of lagged conditional variance. This means that it can be used to model volatility directly as opposed to GJR-GARCH.

3.2.5 Exponential GARCH

The final econometric model that is used in this thesis is the EGARCH model.

$$\ln \sigma^{2t} = \omega + \sum_{i=1}^q \beta_i g(Z_{t-i}) + \sum_{k=1}^p \alpha_k \ln \sigma_{t-k}^2 \quad (3.11)$$

where

$$g(Z_t) = \theta Z_t + \lambda(|Z_t| - E(|Z_t|)) \quad (3.12)$$

where the Z_t is a standard normal variable. The model allows for α and β to be negative, meaning there are no restrictions to finding the parameters. Furthermore, it also models the logarithm of the variance as opposed to the variances modeled in other GARCH iterations. The final addition to the model is the λ which measures the effect of residuals that deviate from the mean.

3.3 Model estimation

While a neural network uses back-propagation to optimize its parameters, a method called *maximum-likelihood estimation* (*MLE*) is used for ordinary GARCH models, and bootstrapping is used for regime-switching GARCH models such as GJR-GARCH and TGARCH.

3.3.1 Log-likelihood estimation

Assume that one has a set of data and wants to find the parameters of the distribution that this data comes from. The probability of getting the data given that they are independently identically distributed can be written as the product of all the probabilities such that,

$$f(x_1, \dots, x_n | \mu, \sigma^2) = \prod_{i=1}^n f(x_i | \mu, \sigma^2) \quad (3.13)$$

The likelihood function used for the approximation of the GARCH parameters is the Gaussian distribution as u_i on the conditional volatility is approximately normal (Hull 2018). Therefore the likelihood function mentioned in Eq. 3.12 becomes,

$$f(x_1, \dots, x_n | \mu, \sigma^2) = \prod_{i=1}^n \left[\frac{1}{\sqrt{2\pi v_i}} \exp\left(\frac{-u_i^2}{2v_i}\right) \right] \quad (3.14)$$

which may be further changed to ease computation by the usage of logarithms, such that

$$\log(f(x_1, \dots, x_n | \mu, \sigma^2)) = \sum_{i=1}^n \left(-\ln v_i - \frac{u_i^2}{v_i} \right) = \ell(\mu, \sigma^2) \quad (3.15)$$

where $u_i = (x_i - \mu)$ and $v_i = \sigma^2$. The final step is to solve for the first order condition such that,

$$\frac{\partial \ell(\mu, \sigma^2)}{\partial \sigma^2} = \frac{\partial \ell(\mu, \sigma^2)}{\partial \mu} = 0 \quad (3.16)$$

which in turn yield the estimators $\hat{\mu}$ and $\hat{\sigma}^2$ used for the model. These are also solved algorithmically using gradient descent similar to back-propagation.

3.4 Model evaluation

When evaluating which models should be used for forecasting, one should use both in-sample and out-of-sample tests on the models' performance. Not only are the models' forecasting capabilities a metric for their viability, but so is the in-sample performance as models can become subject to overfitting. Both of these issues are addressed in the upcoming chapter.

3.4.1 In-sample evaluation

For in-sample model viability, one must adjust the result of the model to the amount of parameters that were used in the estimation. Two widely used criteria are the *Akaike information criterion* (*AIC*) and the *Bayesian information criterion* (*BIC*), where both measure the trade-off between the goodness of fit of the model and the parameter count, rewarding a simpler model over a more complicated one, assuming that the complicated one doesn't provide substantially more explaining power over the simple model.

$$AIC = 2k - 2 \ln \hat{L} \quad (3.17)$$

$$BIC = k \ln n - 2 \ln \hat{L} \quad (3.18)$$

where k denotes the number of parameters in the model, $\hat{L}$ denotes the maximized likelihood function given the chosen parameters and n is the number of data points used as input for the model. The lower the AIC and/or BIC metric is, the better the model. The BIC criteria punishes complex models more than AIC due to the $\ln n$ term as opposed to the linear relationship in the AIC model, $2k$ (Hastie et al. 2009). These two criteria are used to measure how well each GARCH model performs given their parameter count.

3.4.2 Out-of-sample evaluation

For neural networks, the goal is to minimize a certain cost function. This is the same procedure as when you evaluate the parameters of econometric forecasting models (Poon, 2005).

Two common metrics used for model performance are the *root-mean-square percentage error*, ($RMSP\!E$), and the *root-mean-square error* ($RMSE$) where both quantitatively evaluate how far the approximation comes from the realized value.

The *root-mean square error* evaluates the mean squared distance between the forecasts and realized values at each point and takes the square root of said mean such that

$$RMSE = \sqrt{\frac{1}{N} \sum_{t=1}^N (\hat{\sigma}_t - \sigma_t)^2} \quad (3.19)$$

While the $RMSE$ metric is value-dependent, the *root-mean square percentage error* is not, and simply measures the squared difference between the forecast and realized value put in comparison to the realized value

$$RMSP\!E = \sqrt{\frac{1}{N} \sum_{i=1}^N \left(\frac{\hat{\sigma}_t - \sigma_t}{\sigma_t} \right)^2} \quad (3.20)$$

Both evaluation approaches have slightly different properties. The *root-mean square percentage error* is as mentioned a relative measure since it has a normalizing property and can therefore be used when comparing model performance across datasets, which is useful when asserting whether a certain approach works better for one dataset over another. The root-mean-square error on the other hand penalizes large errors disproportionately compared to smaller errors which is suitable in the context of volatility forecasting as a constant and small forecasting error is favorable to higher accuracy but with forecasts that deviate substantially from the realized value. As a result, both of these metrics are used in the analysis of both the GARCH models and the neural networks.

3.4.3 Sliding- or expanding window

A concern when utilizing autoregressive forecasting models is whether the sample window that the model shall use to be trained should be *sliding* or have a fixed start point (*expanding*). It has been found that sliding windows penalize GARCH models while being a good performer for other models (Sahiner, 2022).

4. Data

4.1 Data

The data used for the models will be the daily open, close, high, low, and volume of the OMX Stockholm PI which is the aggregated Swedish stock market, the Polish index WIG, and the Slovenian index SBITOP which were chosen because of their varying degree of market development to identify possible differences in model behaviour between the markets depending on factors such as liquidity. The data starts on the 31st of July 2017 and ends on the 31st of July 2023. The data comes from investing.com. Below in Figure 4.1 and Table 4.1 are the daily returns and summary statistics of the OMXSPI stock index, Figure 4.2 and Table 4.2 depict the daily returns and summary statistics of the WIG index, and finally Figure 4.3 and Table 4.3 show the daily returns and summary statistics of SBITOP between 2017 and 2023.

Table 4.1: Summary statistics of the OMXSPI index

Metric	N	Mean	Median	Std. dev	Kurtosis	Skewness	Min	Max
Returns, %	1507	0.0345%	0.078%	1.197%	7.397	-0.691	-11.135%	7.265%
Volume	1507	$253.8 \cdot 10^6$	$234.6 \cdot 10^6$	$96.7 \cdot 10^6$	8.475	2.088	$52.1 \cdot 10^6$	$1.07 \cdot 10^9$

Table depicting the summary statistics for the percentage daily returns and daily volatility of the OMXSPI index. Data from investing.com.

Figure 4.1: Returns, % of OMXSPI

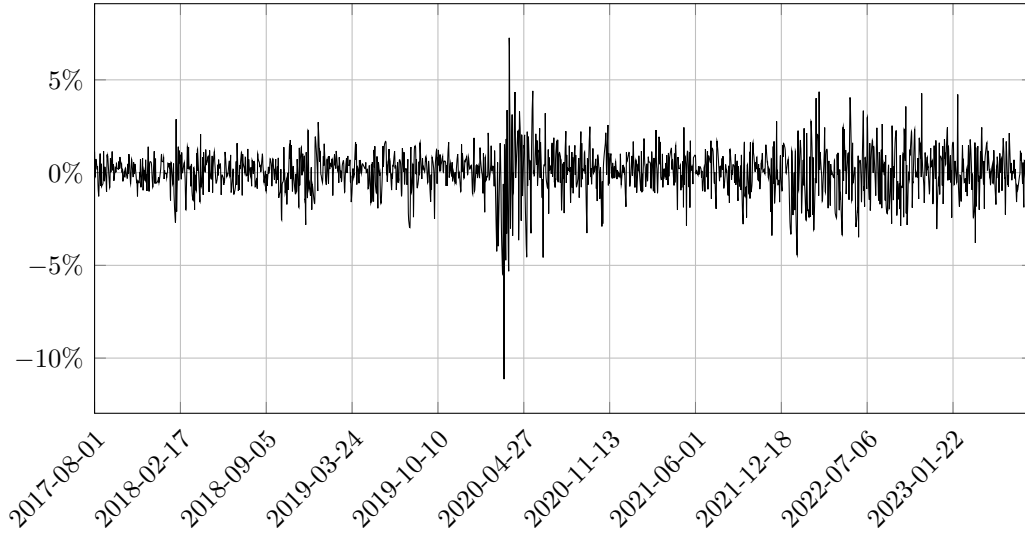


Table 4.2: Summary statistics of the WIG index

Metric	N	Mean	Median	Std. dev	Kurtosis	Skewness	Min	Max
Returns, %	1499	0.0179%	0.0437%	1.286%	11.797	-0.995	-12.652%	7.716%
Volume	1499	$48.3 \cdot 10^6$	$42.9 \cdot 10^6$	$21 \cdot 10^6$	9.746	2.353	$16.5 \cdot 10^6$	$220.2 \cdot 10^6$

Table depicting the summary statistics for the percentage daily returns and daily volatility of the WIG index. Data from investing.com

Figure 4.2: Returns, % of WGI

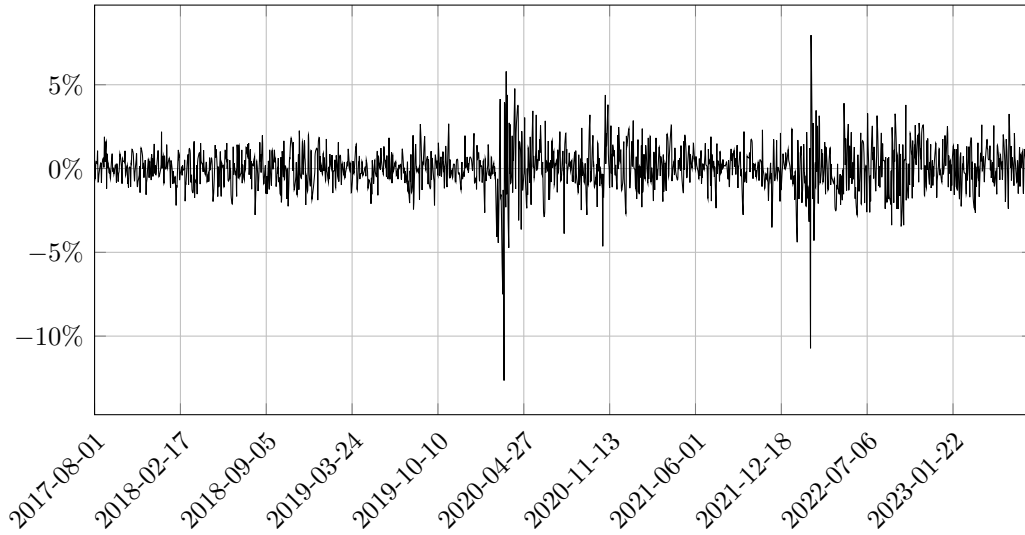
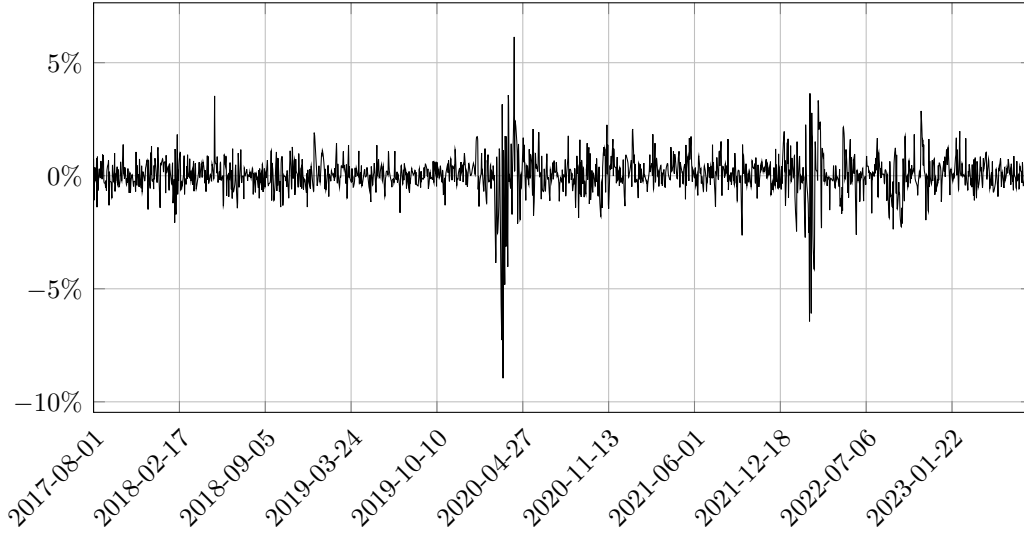


Table 4.3: Summary statistics of the SBITOP index

Metric	N	Mean	Median	Std. dev	Kurtosis	Skewness	Min	Max
Returns, %	1494	0.0331%	0.0426%	0.868%	18.133	-1.589	-8.956%	6.140%
Volume	1494	$23.2 \cdot 10^3$	$18.3 \cdot 10^3$	$21.9 \cdot 10^3$	120.008	8.208	$1.97 \cdot 10^3$	$431.7 \cdot 10^3$

Table depicting the summary statistics for the percentage daily returns and daily volatility of the SBITOP index. Data from [investing.com](https://www.investing.com).

Figure 4.3: Returns, % of SBITOP



4.2 Training set and forecasting technique

The GARCH models will be trained using the entire data set with both a sliding and an expanding sample window. All the data in the three data sets are split up in the following three parts:

- *Training data*: This is the data that the models use to find the appropriate parameters.
- *Validation data*: This is the data that is used to get an unbiased result from the models and to prevent potential overfitting, and lays the foundation of model selection before the testing phase.
- *Test data*: This is the final part of the dataset and is used to find a final optimal model.

The models use 90 days of test data, 252 days of validation data, and the remainder of the data set as training data. The GARCH models are estimated using the Python library

arch with a Student's t distribution using both an expanding window and a sliding window of size 150. The forward-looking 14-day volatility is estimated and is used for the models. The reason for this specific forecasting window is that a window that is too large will have a volatility that will revert towards the long term mean, causing differences in forecasting ability to be small while also being less useful. On the other hand, a window that is too small can be far too unpredictable. The models are updated every day until the last 14-day standard deviation can be calculated, which is on the 14th of July 2023. The neural networks will be trained in a similar fashion, using 90 days of test data and 252 days of validation data using a sliding window, with the network implemented using the Python library **keras**.

The following four neural networks will be used in the analysis

- Single layer with 15 LSTM cells.
- Single layer with 30 LSTM cells.
- Double layer with 30 and 50 LSTM cells respectively.
- Fourth and final model with the same design as the third model except with a final, feed-forward layer of 20 neurons before the output layer.

The neural network models are evaluated based on RMSPE and RMSE similar to the GARCH models and are trained on the 30 past values of the following data

- Logarithmic returns, scaled:

$$100 \cdot (\ln Close_t - \ln Close_{t-1}) \quad (4.1)$$

- Logarithmic trading range:

$$\ln High_t - \ln Low_t \quad (4.2)$$

- Logarithmic trading volume:

$$\ln Volume_t - \ln Volume_{t-1} \quad (4.3)$$

- Past 14-day standard deviation:

$$\sigma(x_{t-13}, x_{t-12}, x_{t-11}, \dots x_t) \quad (4.4)$$

where each variable may be derived from publicly accessible data. These variables were chosen as they can be seen as proxies for volatility. The logarithmic returns were scaled by a factor of 100 for algorithmic purposes since the GARCH models are sensitive to scale, where values above 1 make the GARCH model converge faster. For the neural networks, all the variables were standardized in the domain $[0, 1]$, once again for algorithmic reasons.

4.3 Econometric forecasting

The AIC and BIC tests yielded the lowest values for the Student's t distribution across most of the test data and will therefore be the distribution for the GARCH models. In addition, for each of the GARCH family models, two models were chosen based on their overall lowest AIC and BIC scores among the different parameter options. The following are the models that are used in the analysis, selected based on their AIC and BIC scores: GARCH(1,1), GARCH(1,2), TGARCH(1,1,1), TGARCH(1,1,2), EGARCH(1,1), EGARCH(1,2), GJR-GARCH(1,1,1) and GJR-GARCH(1,1,2). These will be estimated using both a sliding and expanding forecasting window.

5. Results

5.1 Validation forecasts

For the first part of the empirical analysis, models will be chosen based on their results on the validation data. Each data set will be assigned two models from the econometric models and two from the neural network models, resulting in a total of four being used for the final test. The models are chosen based on the metrics RMSPE and RMSE, where the best scores are **bolded** in the tables. The models are chosen independently for each dataset.

Table 5.1: Neural network model validation forecasts for OMXSPI

Model	Metric	
	RMSPE	RMSE
15 cell layered LSTM	0.2171	0.3076
30 cell layered LSTM	0.2386	0.2814
30, 50 cell layered LSTM	0.3654	0.4452
30, 50 cell layered LSTM with 20 feed-forward	0.4330	0.5178

Note: Estimated with the Python library `keras`.

Table 5.2: Econometric model validation forecasts for OMXSPI

Model	Window Type	Metric	
		RMSPE	RMSE
GARCH(1,1)	Sliding	0.3017	0.3276
	Expanding	0.3025	0.3473
GARCH(1,2)	Sliding	0.3069	0.3374
	Expanding	0.3008	0.3435
TGARCH(1,1,1)	Sliding	0.2777	0.2973
	Expanding	0.2267	0.3060
TGARCH(1,1,2)	Sliding	0.2758	0.2970
	Expanding	0.2263	0.3032
GJR-GARCH(1,1,1)	Sliding	0.2694	0.2959
	Expanding	0.2153	0.3047
GJR-GARCH(1,1,2)	Sliding	0.2642	0.2968
	Expanding	0.2124	0.3047
EGARCH(1,1)	Sliding	NaN	NaN
	Expanding	0.2186	0.2867
EGARCH(1,2)	Sliding	NaN	NaN
	Expanding	0.2216	0.2891

Note: Estimated with the Python library `arch`.

Based on this result from the neural network validation data in Table 5.1, a 15 cell layered LSTM and 30 cell layered LSTM should be used for the OMXSPI data set. The econometric models used for the OMXSPI will be the EGARCH(1,1), GJR-GARCH(1,1,2) based on the results in Table 5.2. The reason for this choice of econometric models is that the GJR-GARCH models are superior to EGARCH in terms of RMSPE but are inferior in terms of RMSE. I therefore take the GJR-GARCH with the best RMSPE metric and the EGARCH with the best RMSE metric.

Table 5.3: Neural network model validation forecasts for WIG

Model	Metric	
	RMSPE	RMSE
15 cell layered LSTM	0.3753	0.4395
30 cell layered LSTM	0.3920	0.4570
30, 50 cell layered LSTM	0.2677	0.3374
30, 50 cell layered LSTM with 20 feed-forward	0.3578	0.4115

Note: Estimated with the Python library `keras`.

Table 5.4: Econometric model validation forecasts for WIG

Model	Window Type	Metric	
		RMSE	RMSPE
GARCH(1,1)	Sliding	0.4854	0.5769
	Expanding	0.3708	0.4649
GARCH(1,2)	Sliding	0.4780	0.5647
	Expanding	0.3731	0.4654
TGARCH(1,1,1)	Sliding	0.3215	0.3774
	Expanding	0.3128	0.3814
TGARCH(1,1,2)	Sliding	0.3286	0.3913
	Expanding	0.3227	0.3871
GJR-GARCH(1,1,1)	Sliding	0.3792	0.5016
	Expanding	0.3589	0.4698
GJR-GARCH(1,1,2)	Sliding	0.3942	0.5372
	Expanding	0.3685	0.4748
EGARCH(1,1)	Sliding	NaN	NaN
	Expanding	0.3194	0.3824
EGARCH(1,2)	Sliding	NaN	NaN
	Expanding	0.3269	0.3855

Note: Estimated with the Python library `arch`.

The test portion of WIG will be conducted on the 30 and 50 cell layered LSTM and the 30 and 50 cell with 20 feed forward neurons based on their RMSPE and RMSE values in Table 5.3. From the results above in Table 5.4, the test portion of the WIG dataset will be conducted using TGARCH(1,1,1) with an expanding window and EGARCH(1,1) due to

these two models having performed the best overall

Table 5.5: Neural network model validation forecasts for SBITOP

Model	Metric	
	RMSPE	RMSE
15 cell layered LSTM	0.9530	1.4725
30 cell layered LSTM	0.4209	0.2929
30, 50 cell layered LSTM	0.4705	0.3204
30, 50 cell layered LSTM with 20 feed-forward	0.8319	0.5671

Note: Estimated with the Python library `keras`.

Table 5.6: Econometric model validation forecasts for SBITOP

Model	Window Type	Metric	
		RMSPE	RMSE
GARCH(1,1)	Sliding	1.5134	0.9567
	Expanding	0.4929	0.3185
GARCH(1,2)	Sliding	1.4534	0.9195
	Expanding	0.5011	0.3229
TGARCH(1,1,1)	Sliding	1.2990	0.6956
	Expanding	0.5323	0.3382
TGARCH(1,1,2)	Sliding	1.1355	0.6986
	Expanding	0.5435	0.3441
GJR-GARCH(1,1,1)	Sliding	1.4530	0.9059
	Expanding	0.5257	0.3555
GJR-GARCH(1,1,2)	Sliding	1.4289	0.8921
	Expanding	0.5328	0.3592
EGARCH(1,1)	Sliding	NaN	NaN
	Expanding	0.5477	0.3312
EGARCH(1,2)	Sliding	NaN	NaN
	Expanding	0.5597	0.3379

Note: Estimated with the Python library `arch`.

The validation forecasts in Table 5.5 showed that the most suitable models for the SBITOP are the neural networks containing 30 cells and 30/50 cells with 20 feed forward

neurons. Furthermore, the results from the validation forecasts for the SBITOP index in Table 5.6 show that the two vanilla GARCH models are the best fit for the SBITOP dataset.

5.2 Test forecasts

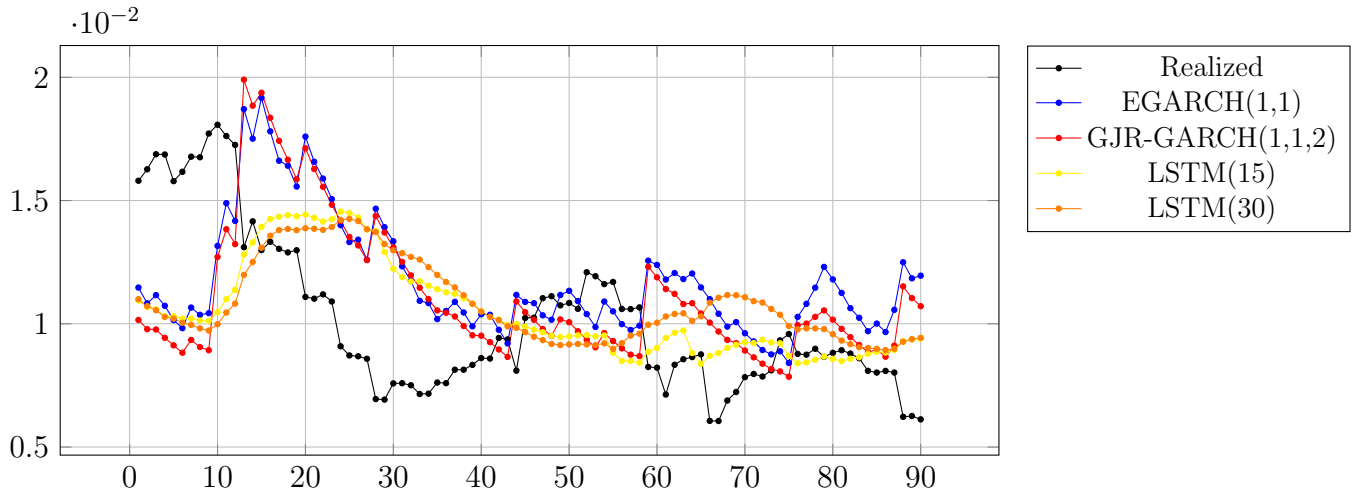
In this section, the models are compared with each other using the last 90 days of the time series that serve as the test data. Each dataset has its econometric and neural network forecasting results in a table followed by a graph depicting them.

Table 5.7: Model test for OMXSPI

Model	Metric	
	RMSPE	RMSE
EGARCH(1,1)	0.4191	0.3797
GJR-GARCH(1,1,2)	0.3941	0.3910
LSTM(15)	0.3371	0.3367
LSTM(30)	0.3858	0.3662

Note: Estimated with the Python libraries `arch` and `keras`.

Figure 5.1: Volatility estimates for OMXSPI depicting 14-day forward volatility on the vertical axis and day on the horizontal axis



The result from the model estimations and training depicted in Table 5.7 is that neural networks are slightly better at forecasting 14-day volatility for this sample of the OMXSPI

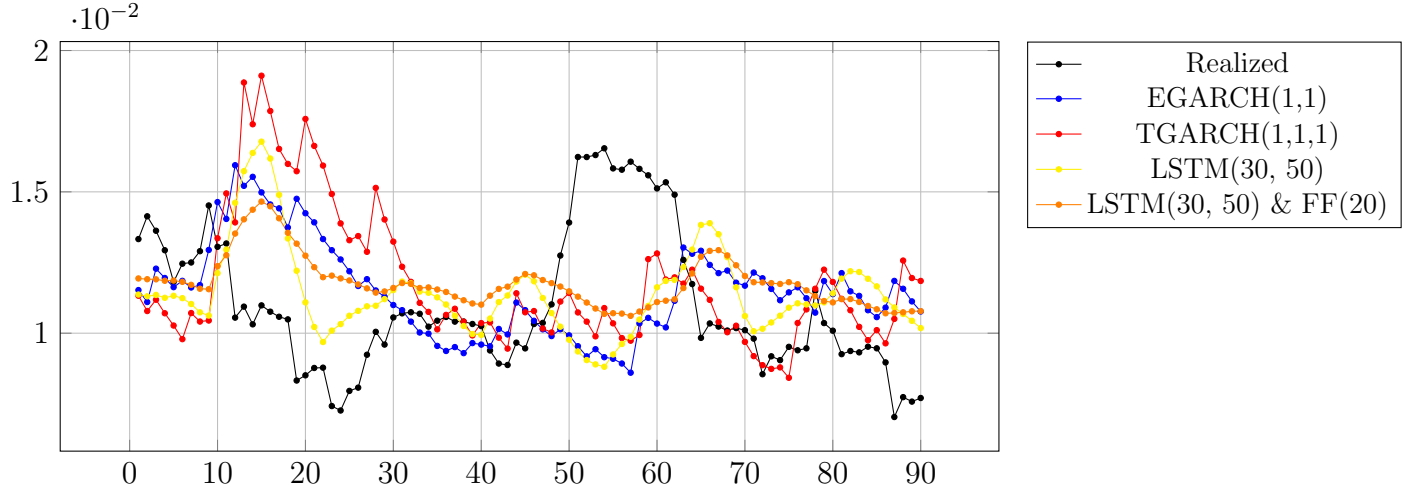
index compared to econometric models, with the 15-neuron layered LSTM providing the best forecast by both RMSPE and RMSE.

Table 5.8: Model test for WIG

Model	Metric	
	RMSPE	RMSE
TGARCH(1,1,1)	0.4239	0.3829
EGARCH(1,1)	0.3143	0.3386
LSTM(30, 50)	0.2702	0.3142
LSTM(30, 50) & FF(20)	0.2725	0.2872

Note: Estimated with the Python libraries `arch` and `keras`.

Figure 5.2: Volatility estimates for WIG depicting 14-day forward volatility on the vertical axis and day on the horizontal axis



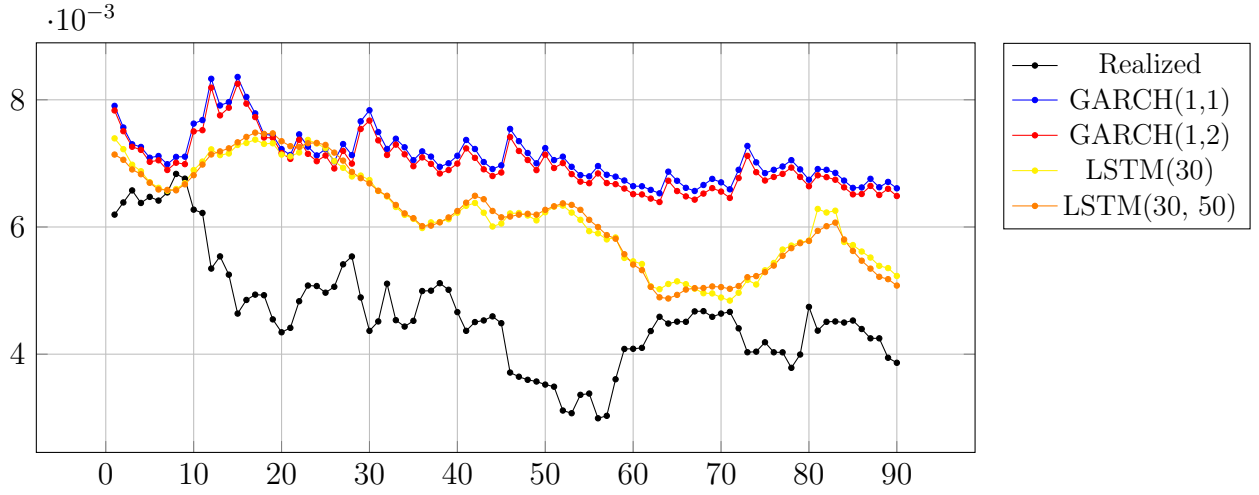
From the estimations on the WIG dataset in Table 5.8 and Figure 5.2, we can also deduce that neural networks perform better overall compared to econometric models. Judging from Figure 5.2, one can see that no model was particularly effective at predicting rapid increases in volatility, however the neural networks, at least to the naked eye, seem to forecast sudden drops in volatility more effectively than EGARCH and TGARCH.

Table 5.9: Model test for SBITOP

Model	Metric	
	RMSPE	RMSE
GARCH(1,1)	0.6411	0.2600
GARCH(1,2)	0.6161	0.2492
LSTM(30)	0.4385	0.1756
LSTM(30, 50)	0.4450	0.1778

Note: Estimated with the Python libraries `arch` and `keras`.

Figure 5.3: Volatility estimates for SBITOP depicting 14-day forward volatility on the vertical axis and day on the horizontal axis



For the final dataset, neural networks once again provided a lower RMSPE and RMSE metric compared to the GARCH models, however, none of the models forecasted volatility particularly well compared to OMXSPI and WIG as can be observed in Table 5.9 as opposed to Table 5.8 and Table 5.7.

6. Conclusion

In this thesis, we have analyzed a host of econometric forecasting models in terms of 14-day forecasting ability. The GARCH, EGARCH, TGARCH, and GJR-GARCH forecasts have been studied on the OMXSPI, WIG, and SBITOP indexes, where the best-performing econometric models have been tested against LSTM neural networks. The variables used in the training of the neural networks have been proxies explored in previous research. The parameters in the GARCH models as well as residual distribution were determined using the Akaike information criterion and Bayesian information criterion ranging from parameter count 1 to 5. Once the econometric models were chosen, their effectiveness was measured using root-mean-square percentage error and root-mean-square error on the validation data where the best models were chosen for the final stage where they were tested on 90 days of test data.

Conclusively, based on the test forecasts done on the indexes OMXSPI, WIG, and SBITOP, one may conclude that neural networks are effective in forecasting volatility in financial markets with their ability to identify non-linearity in data, which is in line with (Bucci 2020) and (Hansson 2017). Another observation is that the econometric models and neural networks performed substantially better for OMXSPI and WIG than for the SBITOP, which is in accordance with (Hansson 2017) who found that the Swedish index had substantially higher predictability for the volatility compared to other indexes.

Furthermore, the predictability of the SBITOP was far lower than the OMXSPI and WIG, while the neural network provided a significantly better model compared to the econometric model. One reason for this result, which would be in line with (Hung, et. al 2019), is that a liquidity metric in the volatility forecasting model improves the accuracy of the forecasts and may indicate that a liquidity metric may disproportionately benefit volatility forecasts for markets where liquidity is low. The incorporation of the logarithmic volume, intra-daily range and realized volatility proposed by (Patton 2011) further improved the accuracy of the neural network volatility forecasts. The validation forecasts in this thesis do however

contradict the findings of (Santos, et. al 2013) and suggest that rolling windows perform better in volatility estimation compared to a fixed window.

6.1 Further research

As mentioned in the delimitations, access to more computing power opens the door to more effective GARCH parameter selection by finding the optimal sliding window and also allows the modeling of more complex neural networks. Furthermore, implementation of other variables such as macroeconomic events into neural networks may lead to models that are better at identifying spikes in volatility.

Another field worth exploring is the introduction of GARCH forecasts in neural networks, where it could be possible to develop more accurate models by intertwining these two approaches.

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A. AIC and BIC results

Table A.1: AIC and BIC for GARCH on OMXSPI with Normal distribution

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	1982.42	1979.1	1980.29	1982.29	1982.87
2	1984.42	1981.1	1981.96	1983.96	1984.81
3	1986.38	1983.09	1980.44	1980.81	1979.8
4	1988.38	1985.09	1982.44	1982.81	1981.8
5	1990.38	1987.09	1984.44	1984.81	1983.8

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	2000.83	2002.11	2007.91	2014.51	2019.69
2	2007.43	2008.71	2014.18	2020.78	2026.23
3	2014.0	2015.31	2017.26	2022.23	2025.83
4	2020.6	2021.91	2023.86	2028.83	2032.43
5	2027.2	2028.52	2030.47	2035.44	2039.03

Table A.2: AIC and BIC for TGARCH on OMXSPI with Normal distribution

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	1953.33	1948.56	1949.36	1951.35	1947.56
2	1954.93	1949.85	1950.63	1952.47	1948.98
3	1953.72	1949.48	1947.97	1949.81	1945.71
4	1955.72	1950.77	1949.97	1951.81	1947.71
5	1957.72	1952.77	1951.97	1954.16	1949.71

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	1976.35	1976.18	1981.58	1988.17	1988.98
2	1982.55	1982.06	1987.45	1993.89	1995.01
3	1985.94	1986.3	1989.39	1995.83	1996.34
4	1992.55	1992.19	1996.0	2002.43	2002.94
5	1999.15	1998.8	2002.6	2009.39	2009.54

Table A.3: AIC and BIC for GJR-GARCH on OMXSPI with Normal distribution

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	1949.42	1945.6	1945.11	1947.11	1942.14
2	1951.42	1947.6	1947.11	1949.11	1944.14
3	1953.3	1949.6	1949.11	1951.11	1946.14
4	1955.2	1951.39	1951.11	1953.11	1948.14
5	1956.81	1953.39	1953.11	1955.11	1950.14

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	1972.43	1973.21	1977.33	1983.93	1983.56
2	1979.04	1979.81	1983.93	1990.54	1990.16
3	1985.52	1986.42	1990.54	1997.14	1996.77
4	1992.02	1992.82	1997.14	2003.74	2003.37
5	1998.23	1999.42	2003.74	2010.34	2009.97

Table A.4: AIC and BIC for EGARCH on OMXSPI with Normal distribution

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	1953.58	1948.42	1949.24	1951.24	1949.45
2	1955.41	1949.99	1950.69	1952.61	1950.78
3	1954.98	1951.09	1943.77	1945.77	1947.13
4	1956.36	1951.75	1945.76	1947.76	1949.02
5	1956.19	1947.36	1941.83	1944.3	1945.32

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	1976.59	1976.03	1981.46	1988.06	1990.87
2	1983.03	1982.21	1987.51	1994.03	1996.81
3	1987.2	1987.91	1985.2	1991.8	1997.76
4	1993.18	1993.17	1991.79	1998.39	2004.26
5	1997.62	1993.39	1992.46	1999.53	2005.15

Table A.5: AIC and BIC for GARCH on OMXSPI with Student's t distribution

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	1959.24	1959.72	1961.69	1963.69	1964.52
2	1961.24	1961.72	1963.55	1965.55	1966.15
3	1963.24	1963.72	1964.24	1965.51	1964.32
4	1965.24	1965.72	1966.24	1967.51	1966.32
5	1967.24	1967.72	1968.24	1969.51	1968.32

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	1982.25	1987.33	1993.91	2000.51	2005.95
2	1988.85	1993.94	2000.37	2006.98	2012.17
3	1995.46	2000.54	2005.66	2011.53	2014.95
4	2002.06	2007.14	2012.26	2018.14	2021.55
5	2008.66	2013.74	2018.86	2024.74	2028.15

Table A.6: AIC and BIC for TGARCH on OMXSPI with Student's t distribution

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	1935.53	1933.68	1935.25	1936.7	1934.87
2	1937.53	1935.63	1937.2	1938.7	1936.87
3	1939.15	1937.37	1938.7	1939.89	1938.1
4	1941.15	1939.14	1940.69	1941.89	1940.1
5	1943.1	1941.14	1942.69	1943.89	1942.1

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	1963.14	1965.89	1972.07	1978.13	1980.89
2	1969.74	1972.45	1978.63	1984.72	1987.5
3	1975.97	1978.79	1984.72	1990.52	1993.33
4	1982.57	1985.16	1991.32	1997.13	1999.93
5	1989.13	1991.77	1997.92	2003.73	2006.53

Table A.7: AIC and BIC for GJR-GARCH on OMXSPI with Student's t distribution

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	1930.66	1929.53	1930.61	1932.61	1929.95
2	1932.66	1931.53	1932.61	1934.68	1931.95
3	1934.66	1933.53	1934.61	1936.61	1933.95
4	1936.66	1935.53	1936.61	1938.61	1935.95
5	1938.66	1937.53	1938.61	1940.61	1937.95

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	1958.28	1961.75	1967.43	1974.04	1975.97
2	1964.88	1968.35	1974.04	1980.7	1982.58
3	1971.48	1974.95	1980.64	1987.24	1989.18
4	1978.09	1981.56	1987.24	1993.84	1995.78
5	1984.69	1988.16	1993.84	2000.45	2002.38

Table A.8: AIC and BIC for EGARCH on OMXSPI with Student's t distribution

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	1935.5	1933.27	1934.87	1936.52	1935.52
2	1937.49	1934.72	1936.24	1937.79	1936.76
3	1938.77	1936.59	1936.6	1937.35	1936.77
4	1940.68	1937.93	1938.54	1939.35	1938.75
5	1942.41	1937.5	1937.83	1938.05	1938.45

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	1963.12	1965.49	1971.69	1977.95	1981.54
2	1969.7	1971.54	1977.66	1983.82	1987.38
3	1975.59	1978.01	1982.62	1987.98	1992.0
4	1982.1	1983.96	1989.17	1994.58	1998.58
5	1988.44	1988.12	1993.06	1997.89	2002.89

Table A.9: AIC and BIC for GARCH on WIG with Normal distribution

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	2089.88	2091.88	2093.88	2095.78	2096.88
2	2091.7	2088.82	2090.82	2092.82	2094.82
3	2093.65	2095.3	2093.82	2096.13	2095.54
4	2095.65	2091.3	2095.82	2093.48	2096.7
5	2097.65	2093.3	2097.82	2099.95	2098.7

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	2108.27	2114.87	2121.46	2127.96	2133.66
2	2114.69	2116.41	2123.0	2129.6	2136.2
3	2121.23	2127.48	2130.6	2137.51	2141.51
4	2127.83	2128.08	2137.19	2139.45	2147.27
5	2134.43	2134.68	2143.79	2150.52	2153.86

Table A.10: AIC and BIC for TGARCH on WIG with Normal distribution

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	2083.41	2083.87	2085.26	2085.18	2083.89
2	2085.41	2089.4	2087.26	2087.15	2085.76
3	2087.41	2091.4	2089.26	2089.15	2087.76
4	2089.41	2090.84	2092.84	2094.52	2089.76
5	2091.41	2092.84	2093.26	2096.52	2091.76

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	2106.39	2111.45	2117.44	2121.96	2125.26
2	2112.99	2121.58	2124.04	2128.53	2131.73
3	2119.59	2128.18	2130.64	2135.12	2138.33
4	2126.19	2132.21	2138.81	2145.08	2144.93
5	2132.78	2138.81	2143.83	2151.68	2151.52

Table A.11: AIC and BIC for GJR-GARCH on WIG with Normal distribution

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	2069.24	2070.48	2072.43	2072.86	2071.76
2	2071.24	2072.48	2074.43	2074.86	2073.76
3	2073.24	2074.48	2076.43	2076.86	2075.76
4	2075.24	2076.48	2078.43	2078.86	2077.76
5	2077.24	2078.48	2080.43	2080.86	2079.76

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	2092.23	2098.06	2104.61	2109.63	2113.14
2	2098.83	2104.66	2111.21	2116.23	2119.74
3	2105.42	2111.26	2117.8	2122.83	2126.33
4	2112.02	2117.86	2124.4	2129.42	2132.93
5	2118.62	2124.45	2131.0	2136.02	2139.53

Table A.12: AIC and BIC for EGARCH on WIG with Normal distribution

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	2072.04	2071.88	2073.68	2075.06	2075.33
2	2074.0	2073.64	2075.45	2076.05	2077.15
3	2074.87	2075.6	2077.44	2077.91	2079.15
4	2073.28	2074.71	2076.13	2077.47	2078.6
5	2074.86	2049.85	2053.21	2051.08	2059.3

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	2095.02	2099.46	2105.86	2111.84	2116.7
2	2101.58	2105.82	2112.23	2117.42	2123.12
3	2107.05	2112.38	2118.81	2123.88	2129.72
4	2110.05	2116.08	2122.11	2128.03	2133.76
5	2116.23	2095.82	2103.78	2106.25	2119.06

Table A.13: AIC and BIC for GARCH on WIG with Student's t distribution

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	2054.12	2056.12	2058.12	2060.11	2061.93
2	2055.43	2051.3	2053.3	2055.3	2062.26
3	2057.43	2053.3	2060.46	2062.14	2063.17
4	2059.43	2054.84	2056.84	2059.09	2061.09
5	2061.43	2056.84	2064.46	2061.09	2063.09

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	2077.11	2083.7	2090.3	2096.89	2103.3
2	2083.01	2083.48	2090.08	2096.68	2108.23
3	2089.61	2090.08	2101.83	2108.11	2113.74
4	2096.21	2096.22	2102.81	2109.66	2116.25
5	2102.81	2102.81	2115.02	2116.25	2122.85

Table A.14: AIC and BIC for TGARCH on WIG with Student's t distribution

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	2047.69	2049.07	2050.43	2051.4	2051.21
2	2049.69	2050.02	2052.02	2053.73	2053.14
3	2051.69	2052.02	2054.02	2055.73	2055.14
4	2053.69	2053.5	2055.5	2057.15	2059.11
5	2055.69	2055.5	2057.5	2059.15	2061.11

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	2075.27	2081.25	2087.21	2092.77	2097.18
2	2081.87	2086.8	2093.39	2099.7	2103.71
3	2088.46	2093.39	2099.99	2106.3	2110.31
4	2095.06	2099.47	2106.07	2112.32	2118.88
5	2101.66	2106.07	2112.67	2118.92	2125.47

Table A.15: AIC and BIC for GJR-GARCH on WIG with Student's t distribution

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	2040.03	2041.87	2043.58	2044.71	2044.88
2	2042.03	2043.87	2045.58	2046.71	2046.88
3	2044.03	2045.87	2047.58	2048.71	2048.88
4	2046.03	2047.87	2049.58	2050.71	2050.88
5	2048.03	2049.87	2051.58	2052.71	2052.88

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	2067.62	2074.05	2080.36	2086.08	2090.85
2	2074.21	2080.64	2086.96	2092.68	2097.45
3	2080.81	2087.24	2093.55	2099.28	2104.05
4	2087.41	2093.84	2100.15	2105.87	2110.65
5	2094.0	2100.43	2106.75	2112.47	2117.24

Table A.16: AIC and BIC for EGARCH on WIG with Student's t distribution

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	2042.4	2043.92	2045.72	2047.06	2048.12
2	2044.34	2044.8	2046.73	2047.77	2049.28
3	2043.59	2045.59	2047.54	2048.64	2050.43
4	2044.29	2046.17	2048.14	2049.82	2051.65
5	2045.49	2043.19	2045.19	2047.19	2049.19

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	2069.98	2076.1	2082.5	2088.43	2094.09
2	2076.52	2081.58	2088.11	2093.74	2099.84
3	2080.37	2086.96	2093.51	2099.21	2105.6
4	2085.67	2092.15	2098.71	2104.99	2111.41
5	2091.46	2093.76	2100.36	2106.96	2113.55

Table A.17: AIC and BIC for GARCH on SBITOP with Normal distribution

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	1487.47	1484.13	1480.57	1482.57	1483.97
2	1489.47	1486.13	1482.48	1484.48	1485.92
3	1491.47	1488.13	1484.48	1486.48	1487.65
4	1493.47	1490.13	1486.48	1488.48	1489.65
5	1495.47	1492.13	1488.48	1490.48	1491.65

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	1505.84	1507.1	1508.13	1514.72	1520.72
2	1512.43	1513.69	1514.64	1521.23	1527.26
3	1519.03	1520.28	1521.23	1527.82	1533.58
4	1525.62	1526.88	1527.82	1534.42	1540.17
5	1532.21	1533.47	1534.42	1541.01	1546.76

Table A.18: AIC and BIC for TGARCH on SBITOP with Normal distribution

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	1491.25	1483.9	1479.77	1481.77	1483.18
2	1493.25	1478.21	1481.77	1483.77	1484.94
3	1495.25	1480.21	1483.77	1485.77	1486.94
4	1499.98	1482.0	1485.77	1479.78	1484.92
5	1501.98	1484.0	1487.77	1481.78	1486.92

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	1514.21	1511.45	1511.92	1518.51	1524.52
2	1520.81	1510.36	1518.51	1525.1	1530.87
3	1527.4	1516.95	1525.1	1531.7	1537.46
4	1536.73	1523.34	1531.7	1530.3	1540.04
5	1543.32	1529.93	1538.29	1536.89	1546.63

Table A.19: AIC and BIC for GJR-GARCH on SBITOP with Normal distribution

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	1477.1	1474.75	1473.35	1475.35	1476.87
2	1479.1	1469.59	1475.35	1472.88	1478.87
3	1481.1	1471.59	1477.35	1479.35	1480.87
4	1483.1	1473.59	1479.35	1476.44	1482.87
5	1485.1	1475.59	1481.35	1478.44	1484.87

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	1500.07	1502.31	1505.5	1512.09	1518.21
2	1506.66	1501.74	1512.09	1514.22	1524.8
3	1513.25	1508.33	1518.68	1525.28	1531.39
4	1519.85	1514.93	1525.28	1526.96	1537.99
5	1526.44	1521.52	1531.87	1533.55	1544.58

Table A.20: AIC and BIC for EGARCH on SBITOP with Normal distribution

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	1489.31	1487.44	1479.51	1481.51	1483.51
2	1482.59	1484.57	1475.71	1477.71	1479.02
3	1484.47	1460.66	1477.08	1479.08	1466.37
4	1483.26	1472.65	1464.47	1465.14	1472.72
5	1481.08	1458.9	1483.55	1462.89	1478.11

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	1512.28	1515.0	1511.66	1518.25	1524.85
2	1510.15	1516.72	1512.45	1519.05	1524.95
3	1516.62	1497.4	1518.42	1525.01	1516.89
4	1520.0	1513.99	1510.41	1515.67	1527.84
5	1522.42	1504.83	1534.07	1518.0	1537.82

Table A.21: AIC and BIC for GARCH on SBITOP with Student's t distribution

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	1382.01	1381.53	1380.3	1382.17	1383.73
2	1384.01	1383.53	1382.3	1384.17	1385.55
3	1386.01	1385.53	1384.3	1386.17	1387.55
4	1388.01	1387.53	1386.3	1388.17	1389.55
5	1390.01	1389.53	1388.3	1390.17	1391.55

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	1404.98	1409.09	1412.45	1418.92	1425.06
2	1411.57	1415.69	1419.04	1425.51	1431.48
3	1418.17	1422.28	1425.64	1432.11	1438.08
4	1424.76	1428.87	1432.23	1438.7	1444.67
5	1431.35	1435.46	1438.82	1445.29	1451.26

Table A.22: AIC and BIC for TGARCH on SBITOP with Student's t distribution

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	1385.15	1384.41	1383.96	1385.96	1387.83
2	1387.15	1384.9	1384.98	1386.98	1388.83
3	1389.15	1388.41	1386.98	1388.98	1390.83
4	1391.15	1388.9	1388.98	1391.16	1393.28
5	1393.15	1390.9	1390.98	1393.16	1395.28

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	1412.71	1416.57	1420.71	1427.3	1433.76
2	1419.3	1421.65	1426.32	1432.91	1439.35
3	1425.9	1429.75	1432.91	1439.51	1445.95
4	1432.49	1434.83	1439.51	1446.27	1452.99
5	1439.08	1441.42	1446.1	1452.87	1459.58

Table A.23: AIC and BIC for GJR-GARCH on WIG with Student's t distribution

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	1379.45	1379.3	1378.4	1380.4	1382.01
2	1381.45	1381.3	1380.4	1382.43	1384.01
3	1383.45	1383.3	1382.43	1384.43	1386.01
4	1385.45	1385.3	1384.4	1386.4	1388.01
5	1387.45	1387.3	1386.4	1388.4	1390.01

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	1407.0	1411.45	1415.15	1421.74	1427.94
2	1413.6	1418.05	1421.74	1428.36	1434.53
3	1420.19	1424.64	1428.36	1434.95	1441.12
4	1426.78	1431.23	1434.93	1441.52	1447.72
5	1433.38	1437.82	1441.52	1448.11	1454.31

Table A.24: AIC and BIC for EGARCH on SBITOP with Student's t distribution

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	1383.65	1383.69	1383.68	1385.68	1387.67
2	1381.95	1383.95	1381.67	1383.67	1385.67
3	1383.88	1375.14	1385.07	1387.07	1389.07
4	1384.63	1384.94	1386.94	1388.94	1390.94
5	1384.75	1386.71	1380.1	1382.1	1384.1

$\downarrow p \quad \vec{q}$	1	2	3	4	5
1	1411.21	1415.84	1420.42	1427.01	1433.6
2	1414.1	1420.7	1423.01	1429.6	1436.2
3	1420.62	1416.48	1431.0	1437.6	1444.19
4	1425.97	1430.87	1437.47	1444.06	1450.65
5	1430.68	1437.23	1435.22	1441.81	1448.41